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Enhancing Operations Quality Improvement through Advanced Data Analytics

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Abstract
This study focuses on the application of data analytics algorithms for real-time monitoring in additive manufacturing processes. The utilization of advanced analytics plays a pivotal role in enhancing the quality control and efficiency of these manufacturing techniques. The research explores how data-driven insights can be harnessed to identify, analyze, and rectify deviations in the manufacturing process, ensuring optimal performance and product quality. By integrating sophisticated monitoring algorithms, the study aims to create a robust framework that continuously analyzes various parameters during additive manufacturing. This includes monitoring factors such as temperature, pressure, and material properties in real-time. The collected data is processed through advanced analytics tools to detect anomalies or deviations from the expected standards. The implementation of machine learning algorithms further facilitates predictive maintenance and proactive adjustments, contributing to the overall reliability and effectiveness of additive manufacturing processes. The outcomes of this research hold significant implications for industries relying on additive manufacturing technologies, providing a foundation for improved process control and product quality. The study contributes to the growing field of Industry 4.0 by showcasing the integration of data analytics as a key enabler for efficient and reliable additive manufacturing.

Keywords- Computer vision, Data analytics, Defect detection, Machine learning, Manufacturing

INTRODUCTION & RELATED WORKS

Additive Manufacturing (AM) is a technology that builds up parts layer by layer using computer-aided design (CAD) and numerical control systems. In recent years, additive manufacturing, commonly known as 3D printing, has emerged as a transformative technology with widespread applications across diverse industries. Despite its remarkable capabilities in creating intricate prototypes and complex geometries, ensuring consistent product quality remains a critical challenge. This study addresses this imperative by proposing an innovative approach that integrates advanced data analytics algorithms into additive manufacturing processes. Positioned within the context of Industry 4.0, the research aims to develop a comprehensive framework for real-time monitoring, analysis, and optimization of key parameters such as temperature, pressure, and material characteristics during the additive manufacturing process. The motivation for this research stems from the need to overcome the inherent complexities of additive manufacturing and to establish tailored solutions for quality control. The primary objectives involve the creation of a data analytics-driven monitoring system, capable of immediate detection of deviations from standard parameters, and the exploration of machine learning algorithms for predictive maintenance, ultimately contributing to enhanced efficiency and reliability in additive manufacturing. The significance of this study lies in its potential to revolutionize additive manufacturing practices, ensuring not only the

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consistent quality of products but also optimizing processes to minimize waste and increase overall efficiency. Molla et al. (2024) discuss the medical textiles sector and how fabric implantation impacts the work in the manufacturing sector, and it is still important [1]. This research builds upon existing literature in both additive manufacturing and data analytics, establishing critical connections between these domains and contributing a novel perspective to the evolving landscape of advanced manufacturing technologies [2, 3]. Over the past two decades, there has been a notable surge in the adoption of Additive Manufacturing (AM) technologies across diverse industries, including aerospace, education, electronics, tool making, automotive, and the medical sector, as highlighted by Brandt et al [4, 5]. Despite the widespread use of AM in part production, the manufacturing process can introduce various defects or anomalies due to multiple factors. Ensuring the integrity of printed parts necessitates the implementation of a robust statistical monitoring or quality control process. Machine Learning (ML) has emerged as a prominent approach for identifying defects or anomalies in AM parts. This involves a meticulous examination of each layer of the part, utilizing machine learning algorithms to discern defective and non-defective layers based on available layer-wise data, as discussed by Sheng et al [6]. Fayshal et al. (2023) interpret what are the risk factors and environmental issues that are associated with human life, and we have planned to consider this thing when considering the quality of products in manufacturing [7, 8]. Ullah et al. (2024) interpret in the paper the cost-effectiveness of using VSM on the production floor which will be the future work for quality controlling in additive manufacturing in our current research and their manufacturing process improvement concept is appreciated [9-12]. Within the realm of metal Additive Manufacturing (AM) technologies, Powder Bed Fusion (PBF) stands out as a method wherein powder material is distributed onto the existing layers using a roller or recoater. This process is facilitated by a reservoir that ensures a continuous supply of fresh material. A heat source, which can be either a laser or an electron beam, is then employed to selectively melt and fuse each layer of metal powder [13]. Among the prevalent Powder Bed Fusion (PBF) technologies in the Additive Manufacturing (AM) industry, Direct Metal Laser Sintering (DMLS) holds a prominent position. This technique enables the printing of parts with a density reaching up to 95%, eliminating the need for supplementary post-build sintering processes [14]. Metal Additive Manufacturing (AM) plays a pivotal role in the manufacturing landscape, particularly in industries such as aerospace, automotive, and medical. In these sectors, AM is driving advancements in innovative designs and applications within a fiercely competitive market. Rahman et al. (2023) and Siddique et al. (2023) work on bitcoin energy saving and cryptocurrency where they have shown the importance of manpower concerning energy work that is a good source of human resources safety-related issues we can consider in our work [15-17]. In aerospace and automotive domains, AM facilitates the development of consolidated and lightweight components, contributing to the creation of more efficient vehicles. Meanwhile, the medical industry leverages a diverse range of high-strength and biocompatible 3D printing materials. This enables manufacturers to customize designs, produce functional prototypes, generate true-to-life anatomical models, and manufacture surgical-grade components [18]. To tackle this challenge, numerous companies have introduced commercial software solutions that offer real-time visualization and monitoring of various process parameters. However, these commercially available solutions often have inherent limitations, primarily in terms of not fully realizing an automated quality control strategy. Additionally, they may fall short in identifying minor defects in the printed part, which could otherwise be automatically rectified before the construction of another layer commences [19]. Rahman et al. (2023) and Ibtisum et al. (2023) analysis of big data tools which is very helpful for our research [20-22]. In recent times, scholars have increasingly employed data analytics methodologies to tackle the challenge of additive manufacturing (AM) monitoring. This predominantly involves integrating offline Machine Learning (ML) techniques with various algorithmic approaches like Computer Vision and signal processing. Notably, recent studies delve into the

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development of monitoring tools tailored for AM machines, striving to incorporate ML into diverse sets of sensed data for automated identification of defects [23-25]. Numerous research endeavours have leveraged machine learning techniques for anomaly detection in 3D printed parts. For instance, Feng et al [26, 27] proposed a machine learning approach for online fault recognition through automatic image processing to detect material defects resulting from process non-conformities in Selective Laser Melting (SLM) of metal powders. This method involved the analysis of in-process images acquired during layer-by-layer SLM processing, employing a bi-stream Deep Convolutional Neural Network-based model. Automated image feature learning and feature fusion were utilized to identify patterns related to SLM defective conditions. Another study focused on using machine learning models to detect geometric defects in additively manufactured objects, where the models were trained on synthetic 3D point clouds with defects and subsequently applied to detect defects in actual production. Jin et al introduced a design for additive manufacturing (AM) to enhance manufacturing efficiency and reduce costs. Mustaquim (2014) describes how mathematically censing data used for land surface temperature is very useful for our research as gradCAM is used in our project for detection purposes [28]. Bazgir et al., (2023) and Molla et al. (2024) used a lot of data using machine learning method for covid related discussion from which we have considered the coding formula for our research with their production method as well [17, 29, 30-32]. They proposed a machine learning integrated design for AM framework, capitalizing on machine learning's capability to discern complex relationships between the design and performance spaces, offering advantages over other surrogate modeling methods due to its ability to model input-output relationships bidirectionally. In the context of Powder Bed Fusion (PBF) AM, a study developed and implemented an in-situ defect detection strategy using supervised machine learning. Achieving defect detection accuracies exceeding 80% during cross-validation experiments, the classifier demonstrated effectiveness after proper training. Addressing the prevalence of defects in the Laser Powder Bed Fusion process, the authors suggested employing computer vision techniques and unsupervised machine learning to differentiate between observed melt pools. By scrutinizing melt pools produced across the process space, in-situ signatures indicative of flaws observed ex-situ were identified. In the present project, the exploration of the dataset commenced with visualizing the relationship between critical factors and their impact on the outcome. Subsequently, multiple machine learning models were constructed to discern between defective and non-defective layers.

METHODOLOGY

Understanding the Parameters of the Datasets

The initial step in the data analysis process involved the importation of the dataset into the Python environment, followed by a thorough exploratory analysis. The preliminary examination focused on checking for the presence of missing values, ensuring data completeness. Given the absence of any missing values, attention was directed towards isolating the variables of interest. A comprehensive 5-number summary was subsequently generated and presented in Table 1. It is imperative to note that the variable "ID bulk Layer" serves as an indicator of whether a layer is deemed defective or not. Specifically, a value of 0 signifies a defective layer, while a value of 1 indicates the opposite condition. This distinction lays the groundwork for further analyses and insights into the dataset.

Table 1: *Five-number summary of good and defective layers.*

	Good Layers			**Defect Layers**		
	x	**Mean LPC**	**Mean SIG**	**x**	**Mean LPC**	**Mean SIG**
Minimum	30338	4485.91	1480.	9559,	4428.63	1422.06
Maximum	31134	4506.15	1559.03	31101	4504.73	1817.29
Mean	30883.87	4494.17	1518.39	16640.24	4458.31	1502.35
Standard deviation	192.89	3.94	15.17	10005.79	27.34,	83.18
Median	30909.0	4493.64	1519.35	9688.5	4443.51	1466.1

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By centring attention on two specific variables, namely Layers and mean SIG, a scatter plot was meticulously crafted to visually represent the intricate interplay between them. This intricately designed scatter plot, elegantly showcased in Fig. 1, serves as an illustrative portrayal of the nuanced relationship between the variables. Notably, the visualization encapsulates the dynamics of these variables across both defective and non-defective layers, offering a comprehensive depiction of their respective behaviours within the dataset. This visual representation aims to provide an insightful and detailed exploration of the correlation between Layers and mean SIG, shedding light on potential patterns or trends that may be discernible in the dataset.

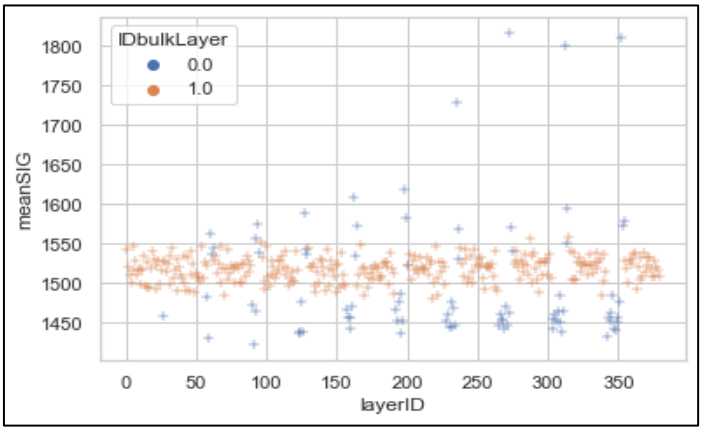

Figure 1: *Layers vs. mean SIG.*

The orange dots on the graphical representation signify the presence of bulk layers, while the blue dots correspond to unexposed layers. For a more detailed presentation of specific information, an additional visualization in the form of a box plot, as depicted in Fig. 2, was meticulously generated. Within this graphical representation, notable observations emerge. The median mean SIG value for unexposed layers hovers around 1460, providing a distinctive contrast to bulk layers, where the median mean SIG value stands at approximately 1520. Going beyond mere medians, the box plot illuminates the distribution of values, showcasing a comparatively narrower range for bulk layers in contrast to the broader span observed for unexposed layers. This nuanced exploration serves to unravel additional insights into the distinct characteristics and statistical variations between these two categories of layers within the dataset. Ahmmed et al. (2023) use deep learning algorithms for brain tumour image processing work, and they considered the confusion matrix significantly from which our detection and confusion matrix development concept is greatly generated [33].

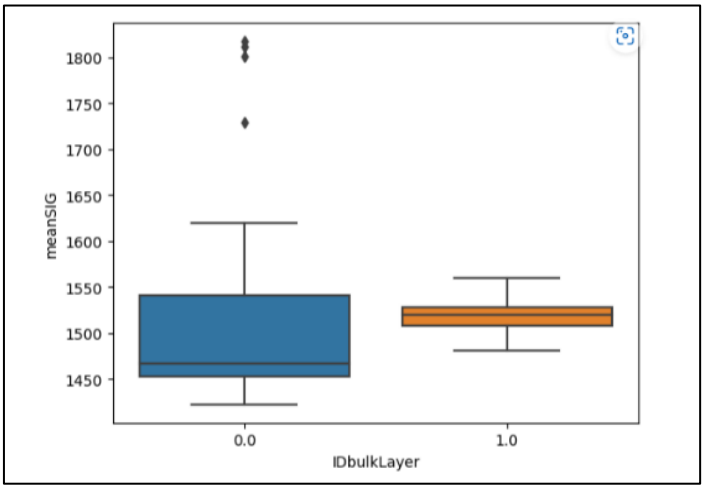

Figure 2: *Boxplot of mean SIG.*

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In the intricately crafted Fig. 3, a scatter plot juxtaposing Layer ID against mean LPC unfolds, providing a compelling visual narrative. A discernible trend emerges, capturing the essence that bulk layers predominantly exhibit elevated mean LPC values when juxtaposed with their unexposed counterparts. The graphical representation masterfully illustrates this distinction, accentuating the propensity of bulk layers to showcase higher mean LPC values. This insightful observation contributes to a more nuanced understanding of the dataset, shedding light on the inherent disparities in mean LPC values between the two categories of layers. The visual depiction encapsulated in Fig. 3 serves as a valuable analytical tool, offering a comprehensive perspective on the interplay between Layer ID and mean LPC within the context of the dataset.

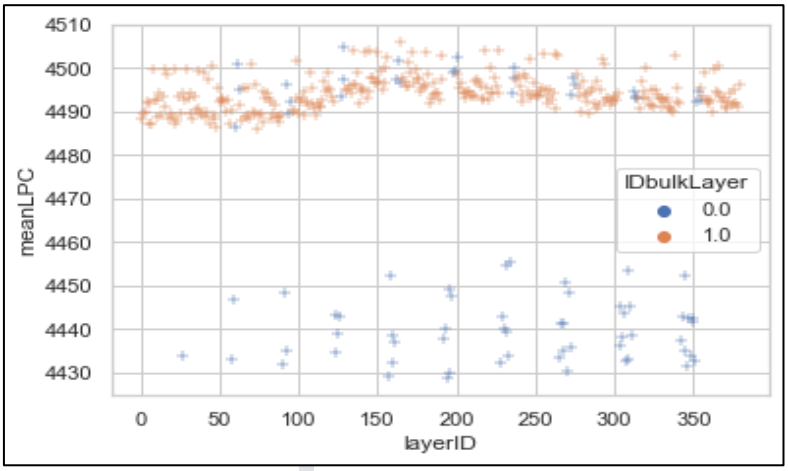

Figure 3: *Layer vs. mean LPC.*

Fig. 4 presents a complementary box plot, enriching our comprehension of the intricate relationship between variables. Within this graphical representation, an insightful juxtaposition unfolds, revealing nuanced details about the distribution of mean LPC values for unexposed and bulk layers. The median mean LPC value for unexposed layers is discerned to be approximately 4443, providing a distinct contrast to bulk layers where the median mean LPC value centres around 4492. Delving beyond medians, the box plot serves as a visual testament to the distributional disparities, with the range of mean LPC values for bulk layers appearing notably more constrained in comparison to the broader span witnessed for unexposed layers. This graphical elucidation in Fig. 4 contributes to a richer understanding of the statistical characteristics inherent in mean LPC values, offering a nuanced perspective on the distinctive traits exhibited by these layers within the dataset.

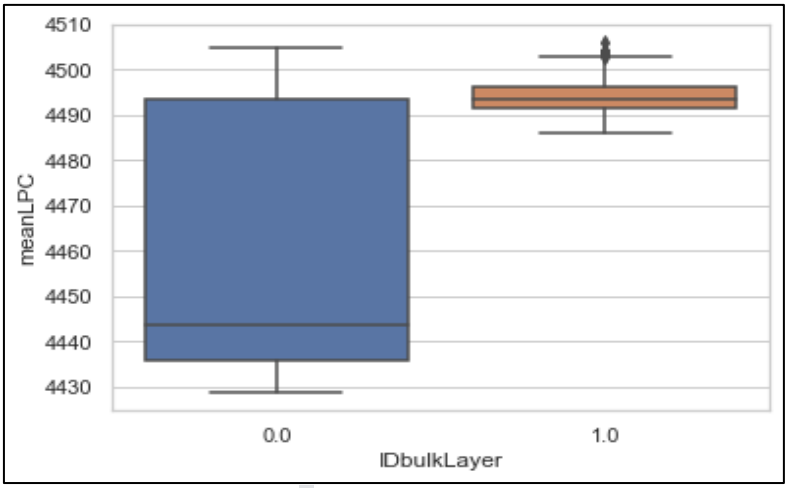

Figure 4: *Layer vs. mean LPC.*

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In the meticulously crafted Fig. 5, a scatter plot depicting the relationship between Layer ID and X unfolds, offering an illuminating visual narrative. Notably, a prominent pattern emerges as the majority of bulk layers exhibit a prevailing trend wherein their associated X values surpass the 30000 thresholds. This observation stands out as a significant and distinctive characteristic, underscoring the inherent differences between bulk layers and unexposed layers within the dataset. While this overarching trend is evident, it's essential to acknowledge the presence of some anomalies or aberrations, highlighting the nuanced nature of the data. The visual insights derived from Fig. 5 contribute to a more comprehensive understanding of the interplay between Layer ID and X, unravelling crucial distinctions that define the unique attributes of bulk layers, with particular attention to their X values. This idea is also taken from this paper [33] as they have discussed smoothly and logically how layer works in deep and transfer learning algorithms.

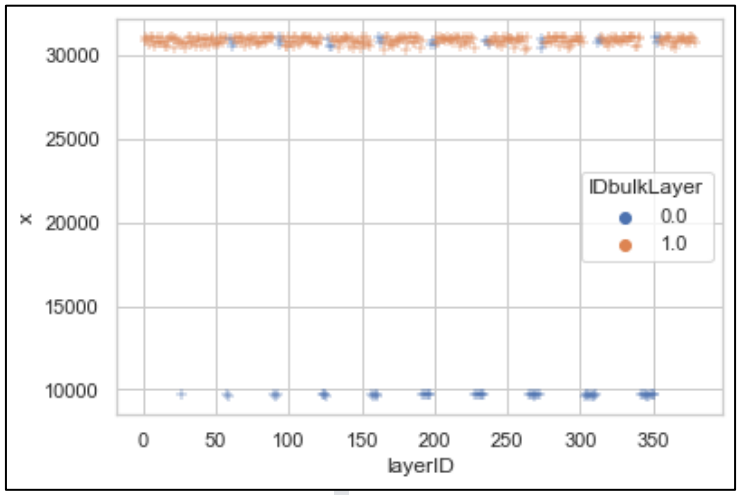

Figure 5: *Layer vs X.*

Upon exhaustive and meticulous scrutiny coupled with the interpretation of the plethora of graphical representations meticulously laid out, a conspicuous realization unfolds. It becomes unequivocally clear that mean SIG, mean LPC, and X stand out as quintessential and highly noteworthy factors that wield considerable influence in the predictive determination of ID bulk Layer. This categorical variable serves as the discriminant between bulk layers and unexposed layers within the dataset. The aggregate insights derived from the diverse graphical depictions coalesce to underscore a coherent and discernible pattern. This pattern, interwoven across mean SIG, means LPC, and X, underscores the profound significance of these specific variables. They not only serve as instrumental elements in unravelling the intrinsic characteristics and disparities residing within the layers of the dataset but also assume a pivotal role in predictive modeling. Consequently, the identified trends and relationships, meticulously teased out from the visual representations, underscore the critical and indispensable role played by mean SIG, mean LPC, and X in the intricate task of predicting the classification of layers as either bulk or unexposed. This comprehensive and holistic analysis not only reaffirms but accentuates the paramount importance of these factors in furnishing a predictive understanding of the ID bulk Layer. Their multifaceted contribution imparts a nuanced and layered context to the interpretation of the dataset, enriching the depth of our insights.

Applying Data Analytics Algorithm

In the initial phase of the procedural approach, the dataset underwent a systematic division, meticulously segregating it into discrete training and testing sections. This segregation process strictly adhered to a proportional distribution, allocating 80% of the data for training purposes and reserving the remaining 20% for testing. This strategic allocation ensures an all-encompassing and thorough representation of each ID bulk Layer

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class within both the training and testing sets. The ensuing elaboration in Table 2 provides an intricate breakdown, offering a detailed account of the specific counts of data points attributed to each category within the delineated training and testing partitions. This methodical partitioning strategy, characterized by its precision and thoroughness, serves as the bedrock for the subsequent phases of robust model training and evaluation. It inherently fosters a judiciously balanced representation of data, meticulously encompassing the diverse classifications present within the ID bulk Layer variable.

Table 2: *Training and testing subset.*

	Train Dataset	Test Dataset	Total
Good layer	237	60	297
Defect layer	65	17	82
total	302	77	379

Metrics derived from testing the machine learning model serve as evaluative benchmarks, offering valuable insights into its performance. Among these metrics, the confusion matrix, precision, recall, and accuracy stand out as pivotal indicators. Accuracy, a fundamental parameter in assessing classification models, represents the percentage of predictions accurately predicted by the model. Precision, another metric, aims to ascertain the percentage of identifications that were accurate. It delves into the precision of the model in correctly identifying positive instances. Simultaneously, recall answers the question of what percentage of actual positives were correctly identified by the model. These metrics collectively provide a comprehensive evaluation framework, shedding light on different aspects of the model's performance and its effectiveness in making accurate predictions.

Linear Regression

For linear regression each of the parameters is individually considered to train the ML model and then test it Please find the attached snapshots below in Fig. 6:

Multiple Linear Regressions

A machine learning algorithm called linear regression is based on the supervised regression technique. Regression uses independent variables to model a goal prediction value. It is mostly used to determine how variables and forecasting relate to one another. Regression models vary according to the number of independent variables they use as well as the type of relationship they consider between the dependent and independent variables. By using the pre-formulated logistic regression algorithm from the sklearn package in Python, import Logistic Regression was possible. Then, we created three separate ML models, one for mean SIG, mean LPC, and x, respectively. Firstly, for mean SIG an ML model is constructed to predict the good or defective layers.

```
#for parameter meanSIG
X_train = np.array(training_set['meanSIG']).reshape(-1, 1)
y_train = np.array(training_set['IDbulkLayer']).reshape(-1, 1)
X_test = np.array(testing_set['meanSIG']).reshape(-1, 1)
y_test = np.array(testing_set['IDbulkLayer']).reshape(-1, 1)

regr = LogisticRegression()
regr.fit(X_train, y_train)
y_pred=regr.predict(X_test)
print(metrics.classification_report(y_pred, y_test))
```

```
              precision    recall  f1-score   support

         0.0       0.00      0.00      0.00         0
         1.0       1.00      0.78      0.88        77

    accuracy                           0.78        77
   macro avg       0.50      0.39      0.44        77
weighted avg       1.00      0.78      0.88        77
```

Figure 6: *Linear regression.*

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Here we can see that the accuracy is 78% additionally linear regression doesn't provide any confusion matrix. Not ideal for classification-type problems.

Multi-Variable Logistic Regression Model

Logistic regression operates as a supervised classification algorithm, particularly effective in scenarios where the target variable (output), denoted as y, assumes discrete values within a given set of features (inputs), represented by X. In the context of this project, the training set involves the exclusion of the target variable ID bulk layer, while the testing set exclusively comprises the target variable (Y). Leveraging the Python sklearn library, the multinomial logistic regression model is applied, utilizing the newton-cg solver. The ensuing section delves into the metrics and confusion matrix of the machine learning model, intricately presented in Fig. 7 for comprehensive visualization and analysis.

```
              precision    recall  f1-score   support

           0       0.94      1.00      0.97        16
           1       1.00      0.98      0.99        61

    accuracy                           0.99        77
   macro avg       0.97      0.99      0.98        77
weighted avg       0.99      0.99      0.99        77

confusion matrix
[[16  1]
 [ 0 60]]
```

Figure 7: *Metrics of logistic regression model.*

In the analysis of the results, we can derive meaningful insights from the accuracy metric, which pertains to both classes of ID bulk layer, distinguishing between a good or defective layer. The remarkable accuracy value of 0.99 signifies that the ML model consistently predicts the layers correctly, almost without exception. Additionally, an examination of the confusion matrix reveals a minimal occurrence of only one false positive, further accentuating the model's exceptional predictive accuracy.

Decision Tree

The decision-making process is commonly visualized through decision trees, characterized by a branching, tree-like structure. This visual flowchart is frequently employed to strategize and chart both business and operational decisions. In the outlined methodology, decisions unfold and lead to consequences, giving rise to a tree-like structure or graphical representation. In the realm of machine learning, decision trees serve as a pivotal method for structuring algorithms. This involves segmenting the dataset features utilizing a decision tree technique and a cost function. Before optimization, the decision tree undergoes expansion, with the removal of branches that would incorporate redundant features, a process known as pruning. Additionally, adjustments to parameters like the decision tree's depth are made to mitigate the risk of overfitting or the creation of an excessively intricate tree. Notably, Python's sklearn library encompasses a built-in decision tree classifier algorithm. This algorithm takes inputs X (independent variables) and Y (dependent variable), with Y representing the target variable ID bulk layer, excluded from both the X train and X test. Subsequently, the algorithm is employed to train the machine learning model with criterion = entropy and default max_depth value. The model is then tested to obtain metrics, as illustrated in the figure below (Fig. 8).

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precision, and recall Show the confusion matrix

```
from sklearn.tree import DecisionTreeClassifier
X_train = training_set.drop('IDbulkLayer', axis=1)
y_train = np.array(training_set['IDbulkLayer']).reshape(-1, 1)
X_test = test_set.drop('IDbulkLayer', axis=1)
y_test = np.array(test_set['IDbulkLayer']).reshape(-1, 1)
#decision tree with entropy and default max_depth
model = DecisionTreeClassifier(criterion='entropy', max_depth=None)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(metrics.classification_report(y_pred, y_test))
print("confusion matrix")
cm = metrics.confusion_matrix(y_test, y_pred)
print(cm)
```

```
              precision    recall  f1-score   support

         0.0       0.94      0.94      0.94        17
         1.0       0.98      0.98      0.98        60

    accuracy                           0.97        77
   macro avg       0.96      0.96      0.96        77
weighted avg       0.97      0.97      0.97        77

confusion matrix
[[16  1]
 [ 1 59]]
```

Figure 8: *Decision tree with default max_depth.*

In the analysis of the model's performance, it is evident that the accuracy stands at 97%. A closer examination of the confusion matrix reveals that the model achieved 16 true positives, accurately predicting instances of actual positives. Furthermore, 59 true negatives were correctly identified by the model. Interestingly, both false positives and false negatives amount to 1 each, indicating instances where the model predicted positive and negative outcomes, and vice versa. This nuanced interpretation underscores the model's ability to effectively discern between positive and negative classifications while acknowledging the minor instances of misclassification (Fig. 9).

```
              precision    recall  f1-score   support

           0       0.94      1.00      0.97        16
           1       1.00      0.98      0.99        61

    accuracy                           0.99        77
   macro avg       0.97      0.99      0.98        77
weighted avg       0.99      0.99      0.99        77

confusion matrix
[[16  1]
 [ 0 60]]
```

Figure 9: *Metrics for max_depth=4 with Decision tree.*

Here we can see an improvement in the confusion matrix. False negative is now 0 and thus accuracy also improved. Now based on the accuracy a bar plot is plotted in Fig. 10.

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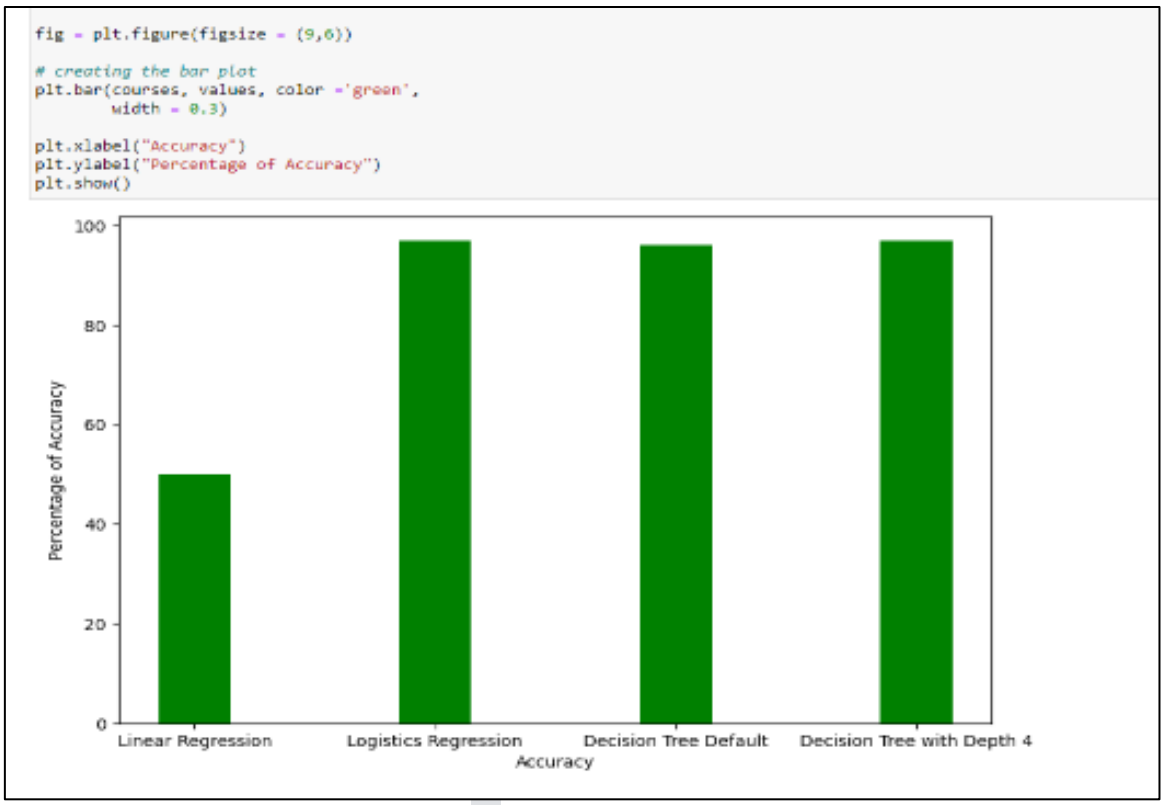

Figure 10: *Bar chart of accuracy.*

Upon scrutinizing the outcomes of the various algorithms, it becomes evident that the linear regression model exhibits the lowest accuracy. This is attributed to its inherent unsuitability for classification-type problems, lacking the capability to provide distinct class categories. In contrast, logistic regression stands out with a notably high accuracy. This is attributed to several factors, including its capacity to make no assumptions about class distributions in feature space, its adaptability for multiple classes (multinomial regression), and its natural probabilistic perspective on class predictions. Logistic regression also demonstrates a lower susceptibility to overfitting, although caution is warranted in high-dimensional datasets where overfitting can occur. Regularization techniques, such as L1 and L2 regularization, can be employed to mitigate overfitting in such scenarios. Moving on to the decision tree algorithm, it is observed that the default max depth yields an accuracy of 0.97. Notably, when adjusting the max_depth to 4, the accuracy surges to 0.99. This nuanced analysis underscores the impact of parameter tuning on the model's performance, emphasizing the importance of optimizing hyperparameters to enhance accuracy in specific contexts.

CONCLUSION

This study aims to develop machine learning models specifically designed to differentiate between layers classified as either defective or non-defective within the additive manufacturing process. The inherent challenge arises from the layer-by-layer manufacturing approach, which exhibits minimal tolerance for variations in product quality. Recognizing the potential resource efficiency gained through early identification of flawed layers, this research seeks to establish a statistical model that utilizes a range of parameters. The project addresses this imperative by introducing a machine-learning algorithm crafted to detect defective layers at an early stage in the manufacturing process. While multiple machine learning models were developed for comparative analysis in this study, significant emphasis was placed on decision tree models (max_depth = 4) and logistic regression models, identified as the most effective in discerning between defective and non-defective layers. Computer Vision and Machine Learning have emerged as promising solutions for addressing the in-situ monitoring challenges within the Additive Manufacturing (AM) process. Two key challenges motivating

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these approaches include the absence of an effective data collection infrastructure tailored specifically to AM and the requirement to train models on large, annotated datasets, which are often costly and challenging to acquire in industrial settings. This paper introduces an economical, camera-based, in-situ defects monitoring system designed for metal Powder Bed Fusion (PBF). The initial prototype, created and tested within a genuine industrial setting at an automotive company (FCA Product Development AM Centre), incorporates real-time Computer Vision and Machine Learning algorithms. These algorithms aim to identify five distinct categories of powder bed defects and enable layer-wise monitoring of a printed part's profile. The preliminary findings indicate that the algorithms exhibit commendable performance concerning defect detection accuracy and profile segmentation. Furthermore, they demonstrate suitability for real-time execution using cost-effective hardware.

FUTURE WORK

The exploration of more sophisticated machine learning models and algorithms presents a promising direction. Investigating the integration of deep learning techniques, such as neural networks, could enhance the predictive capabilities of analytics systems, allowing for a more nuanced understanding of complex operational processes. Additionally, the development of real-time analytics frameworks should be considered to enable instantaneous decision-making. Implementing advanced algorithms that can process and analyze data in real-time would empower organizations to respond promptly to evolving operational scenarios, reducing the likelihood of defects or inefficiencies. Another important aspect of future research involves the incorporation of data from diverse sources. Expanding the scope of data analytics to encompass information from Internet of Things (IoT) devices, sensors, and external databases can provide a more comprehensive understanding of operational dynamics. This holistic approach would contribute to a more accurate identification of potential areas for improvement and facilitate a proactive rather than reactive approach to quality enhancement. Furthermore, the study of interpretability and explainability in data analytics models is crucial for fostering trust and adoption in operational settings. Future research could delve into creating transparent and interpretable models, enabling decision-makers to understand the rationale behind recommendations or predictions. This is especially pertinent in industries where operational decisions have significant consequences. Considering the dynamic nature of industries, research could explore adaptive analytics systems. Investigating methodologies that allow data analytics models to self-adjust and evolve based on changing operational conditions would be instrumental. This adaptive capability could enhance the resilience of operations and ensure continuous quality improvement in the face of evolving challenges. In summary, future work in the field should aim to advance the sophistication of analytics models, enable real-time decision-making, incorporate diverse data sources, enhance model interpretability, and explore adaptive analytics systems. These directions hold the potential to significantly contribute to the ongoing efforts of enhancing operational quality through advanced data analytics.

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